



THE UNIVERSITY
OF QUEENSLAND
AUSTRALIA

Student Number Family Name First Name

Do not distribute this document

School of Mathematics & Physics
EXAMINATION

Semester One Final Examinations, 2020

MATH1061 Discrete Mathematics

This paper is for St Lucia Campus students.

Examination Duration: 120 minutes (+ additional 30 mins encompassing reading time, and time to scan and upload solutions). You must commence your exam at the time listed in your personalised timetable. The exam will remain open **only** for the duration shown.

Materials Permitted While Completing the Exam: You may use any online or physical material you wish. However, you have to write solutions in your own words (i.e. copying is considered cheating). You are NOT allowed to contact another human being directly or indirectly regarding this test. In particular, posting questions to chat forums is considered cheating.

Instructions to Students: Answer all questions. Upload as a **SINGLE PDF FILE**. You can print the exam and write on the exam paper, or write your answers on blank paper, or write electronically on a suitable device. Scan or photograph your work if necessary and upload your answers to Blackboard as a single pdf file.

Who to Contact Since students may not all undertake the online exam at the same time, or in the same time zone, and that some questions may be randomised, responding to student queries and/or relaying corrections to exam content during the exam will not be feasible. Course coordinators will not be able to respond to academic queries during the exam.

If you have any concerns or queries about a particular question, or need to make any assumptions to answer the question, state these at the start of your solution to that question. You may also include queries you may have made with respect to a particular question, should you have been able to 'raise your hand' in an examination room.

If you experience any technical difficulties during the exam, contact the [Library AskUs](https://web.library.uq.edu.au/contact-us) service via the Live Chat or Phone for advice (<https://web.library.uq.edu.au/contact-us> - open 7:00am – 10:00pm AEST every day during the final exam period). You should ask the library staff for an email documenting the advice provided so you can provide this to the course coordinator.

Certification (must be signed before submission):

I certify that my submitted answers are entirely my own work and that I have neither given nor received any unauthorised assistance on this assessment item.

Signed: _____ Date: _____

1. Let the cyclic group $\{[0], [1], [2], \dots, [n-1]\}$ be denoted by $\mathbb{Z}/n\mathbb{Z}$. Consider the following statement: for every positive integer n and every x in $\mathbb{Z}/n\mathbb{Z}$, there exists $y \in \mathbb{Z}/n\mathbb{Z}$ such that $xy = [1]$.

(a) (2 marks) Write the negation of this statement.

(b) (2 marks) Is the original statement true or false? Justify your answer.

2. (4 marks) Let x be a fixed positive integer. Is it possible to have a graph G with $4x + 1$ vertices such that G has a vertex of degree d for all $d = 1, 2, \dots, 4x + 1$? Justify your answer.
(Note: The graph G does not need to be simple.)

3. (4 marks) Let q be an odd prime. Let x be an integer satisfying $1 < x < q$. Prove that

$$q \mid \binom{q}{x}.$$

4. (4 marks) Consider the function $f : \mathbb{R} - \{\pm 1\} \rightarrow \mathbb{R}$ defined by

$$f(x) = \frac{1}{x^2 - 1}.$$

Determine the range of f . Justify your answer.

5. (4 marks) Let x be a fixed but arbitrary integer. Do there exist infinitely many integers y such that

$$y^3 \equiv x^{12} \pmod{2019}?$$

Justify your answer.

6. Let S be a finite set and let $\mathcal{P}(S)$ denote the set of all subsets of S . Define a relation on $\mathcal{P}(S)$ by declaring that two subsets A and B are related if A and B have the same number of elements.

(a) (2 marks) Prove that this is an equivalence relation.

(b) (2 marks) Determine the equivalence classes.

(c) (2 marks) Determine the number of elements in each equivalence class.

7. (6 marks) Consider the sequence $\{a_n\}_{n \geq 1}$ defined by $a_1 = 3$, $a_2 = 8$, and

$$a_n = 4(a_{n-1} - a_{n-2}), \quad \forall n \geq 3.$$

Prove that $a_n = 2^{n-1}(n+2)$ for all $n \geq 1$.

8. (8 marks) A group of 3 friends go to a concert with total attendance 30. At the end of the concert, the coat check system crashes, so they return people's coats at random. Determine the probability that at least one of the 3 friends obtains their own coat. Write your answer in the fraction form $\frac{a}{b}$ where $\gcd(a, b) = 1$.

9. (10 marks) Define a binary operation on $\mathbb{R}^2 - \{(0, 0)\}$ by

$$(a, b) \cdot (c, d) = (ac - bd, ad + bc).$$

Prove that $(\mathbb{R}^2 - \{0\}, \cdot)$ is an abelian group.

(You do not need to prove that the operation is closed.)

10. (10 marks) Let x be an integer bigger than 4. Prove that x is prime if and only if x does not divide $(x-1)!$.

11. (10 marks) The AFL (Australian Football League) has 18 teams. Each team plays one game per week. Assume a team never plays a given other team more than once. Prove that after 8 weeks, there exists at least 3 teams none of which have played each other.